



ARI2129 - Principles of CV

Gamma Correction & Tone Mapping

Alexander Busuttil Cazac, Nicholai Camilleri, Sarah Demanuele,
Zack Haber & Reuben Mifsud



Introduction to Gamma Correction

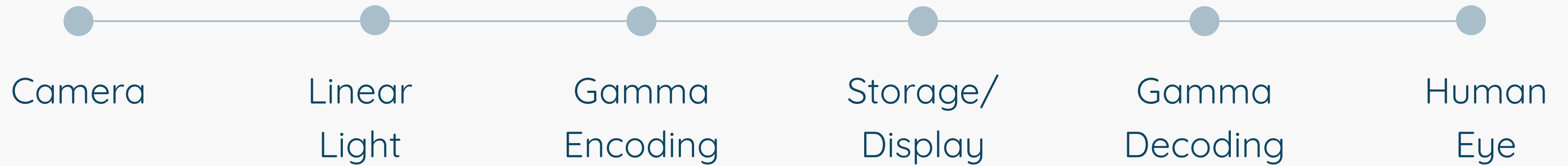
- CRT Displays
- Near-Logarithmic Perception of Light



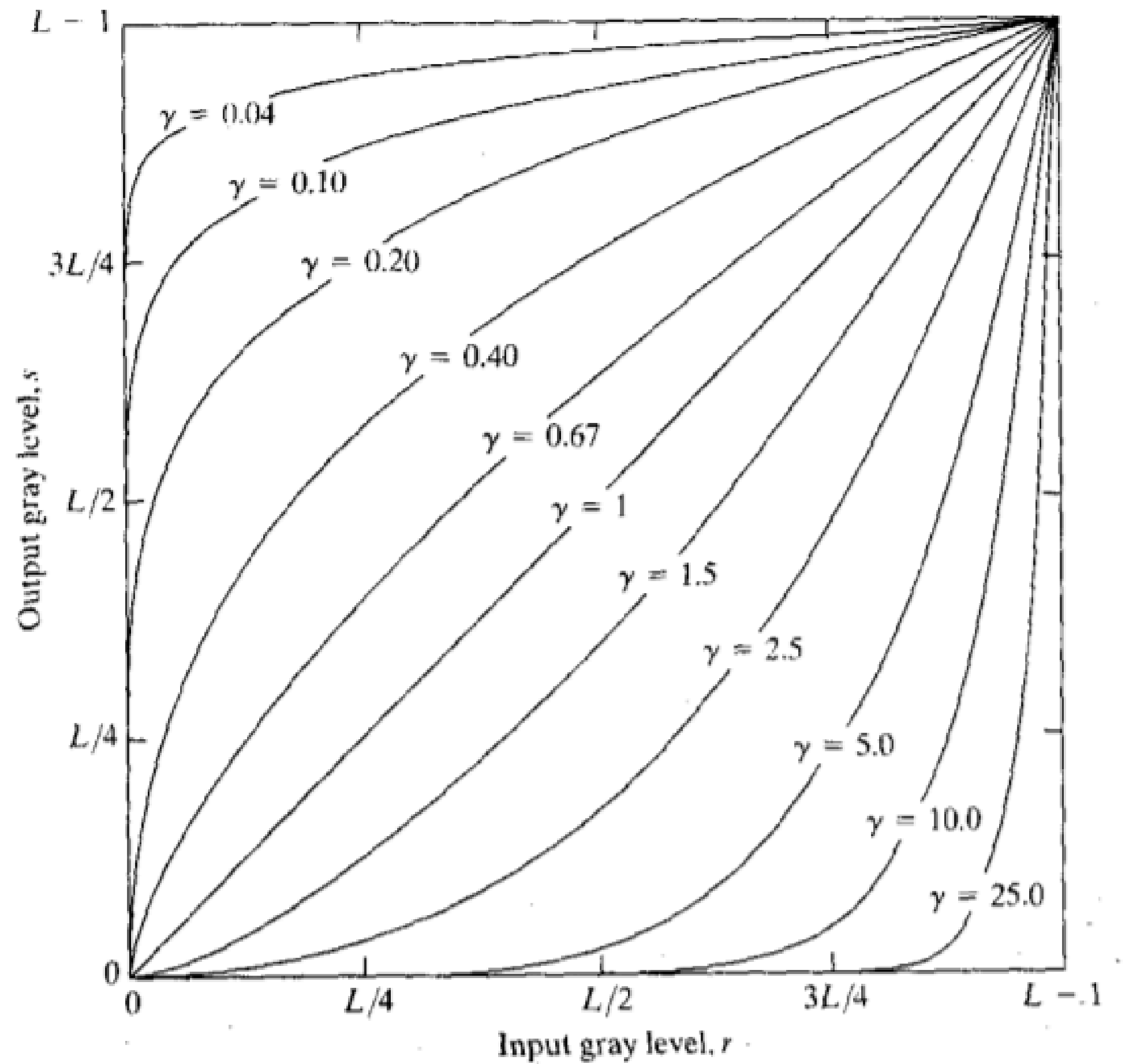
“If cameras store light linearly, but our eyes perceive it logarithmically, how do we store an image that actually looks natural to a human?”



The Solution



$$s = C \cdot r^\gamma$$



Original



Gamma = 1.5



Gamma = 0.57



Gamma = 0.4



SIMULATOR

Introduction to Tone Mapping

Dynamic Range

1

The span between the darkest and brightest values a system can simultaneously capture or display

Low Dynamic Range

2

An image format with a limited brightness range

High Dynamic Range

3

An image format that stores a wider range of brightness values



LDR



HDR



“If an image was captured in HDR, how can we display it on an LDR screen while maintaining the overall quality of the original image?”



Tone Mapping

What?

Remapping HDR (thousands of nits) → LDR (~100 nits) while preserving perceived detail

Why?

- Clipping
- Crushing

Tone mapping prevents both

How?

A Tone Mapping Operator (TMO) applies a non-linear curve to luminance values:

- Preserves detail
- Compresses highlights without clipping

Common operators: Reinhard · ACES Filmic · Hable (Uncharted 2)



SIMULATOR

Real-World Applications

- 01** Photography & Camera
- 02** Advertising & Marketing
- 03** Radiographic Imaging
- 04** Video Games
- 05** Display Hardware

Limitations & Failure Cases

	Problem	When	Effect
01	blurry edge detection	local tone mapping	glowing edges
02	double gamma	post-production	washed-out image
03	dark areas amplified	HDR video	grain in dark areas
04	colour desaturation	global operators	unrealistic colours

When not to use

Gamma Correction

- Mid-pipeline rendering
- Image already gamma encoded

Tone Mapping

- Content is already SDR
- If the display natively supports the full HDR range

Alternatives

Gamma Correction

- Perceptual quantiser
- Linear workflow with colour management

Tone Mapping

- Exposure fusion
- Local adaptation with spatial awareness